

Tutorial-3.1&3.2

2. Suppose that a life insurance company has guaranteed a payment of \$14 million to a pension fund 4.5 years from now. If the life insurance company receives a premium of \$10.4 million from the pension fund and can invest the entire premium for 4.5 years at an annual interest rate of 6.25%, will it have sufficient funds from this investment to meet the \$14 million obligation?

3. Answer the below questions.

(a) The portfolio manager of a tax-exempt fund is considering investing \$500,000 in a debt instrument that pays an annual interest rate of 5.7% for four years. At the end of four years, the portfolio manager plans to reinvest the proceeds for three more years and expects that for the three-year period, an annual interest rate of 7.2% can be earned. What is the future value of this investment?

(b) Suppose that the portfolio manager in Question 3, part a, has the opportunity to invest the \$500,000 for seven years in a debt obligation that promises to pay an annual interest rate of 6.1% compounded semiannually. Is this investment alternative more attractive than the one in Question 3, part a?

4. Suppose that a portfolio manager purchases \$10 million of par value of an eight-year bond that has a coupon rate of 7% and pays interest once per year. The first annual coupon payment will be made one year from now. How much will the portfolio manager have if she (1) holds the bond until it matures eight years from now, and (2) can reinvest all the annual interest payments at an annual interest rate of 6.2%?

7. Consider a bond selling at par (\$100) with a coupon rate of 6% and 10 years to maturity.

(a) What is the price of this bond if the required yield is 15%?

(b) What is the price of this bond if the required yield increases from 15% to 16%, and by what percentage did the price of this bond change?

(c) What is the price of this bond if the required yield is 5%?

(d) What is the price of this bond if the required yield increases from 5% to 6%, and by what percentage did the price of this bond change?

(e) From your answers to Question 7, parts b and d, what can you say about the relative price volatility of a bond in a high-interest-rate environment compared to a low-interest-rate environment?